CS471 – Web Technologies (Laboratory)	Qassim University College of Computer	Lab 9
		Django Models (Part 3)

This lab session covers Django Models (Part 3). By the end of this session, students should have a very good understanding of how to make advanced queries using annotations.

Pre-lab Preparation:

1. The following DB models scheme

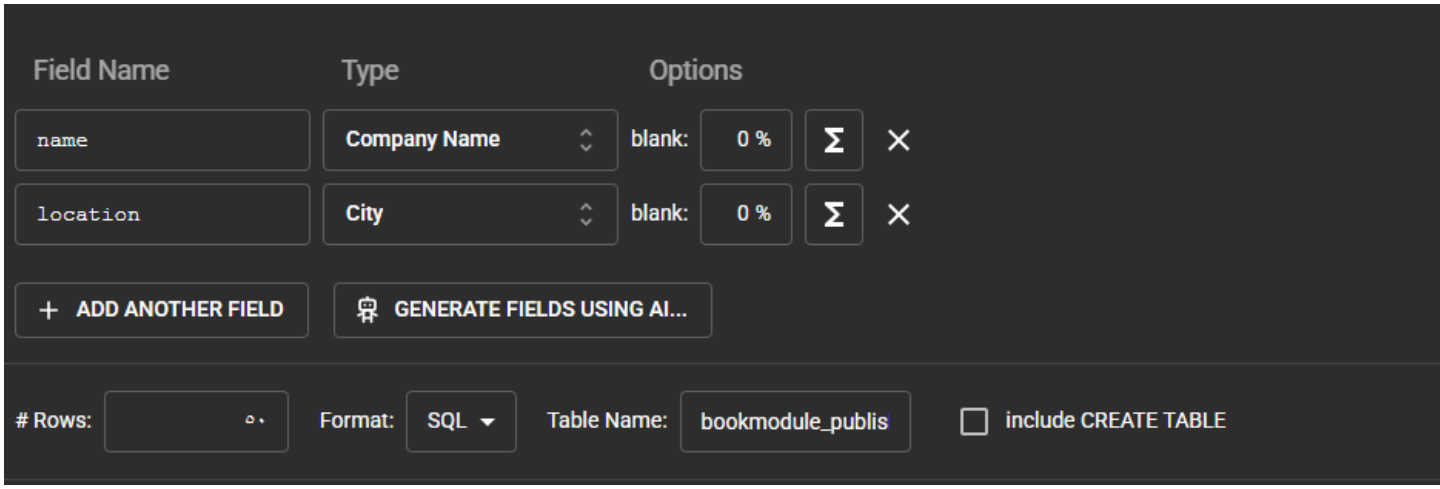
```
class Publisher(models.Model):
  name = models.CharField(max_length=200)
  location = models.CharField(max_length=300)

class Author(models.Model):
  name = models.CharField(max_length=200)
  DOB = models.DateField(null=True)

class Book(models.Model):
  title = models.CharField(max_length= 100)
  price = models.FloatField(default=0.0)
  quantity = models.IntegerField(default=1)
  pubdate = models.DateTimeField()
  rating = models.SmallIntegerField(default = 1)

  publisher = models.ForeignKey(Publisher, null=True, on_delete=models.SET_NULL)
  authors = models.ManyToManyField(Author)
```

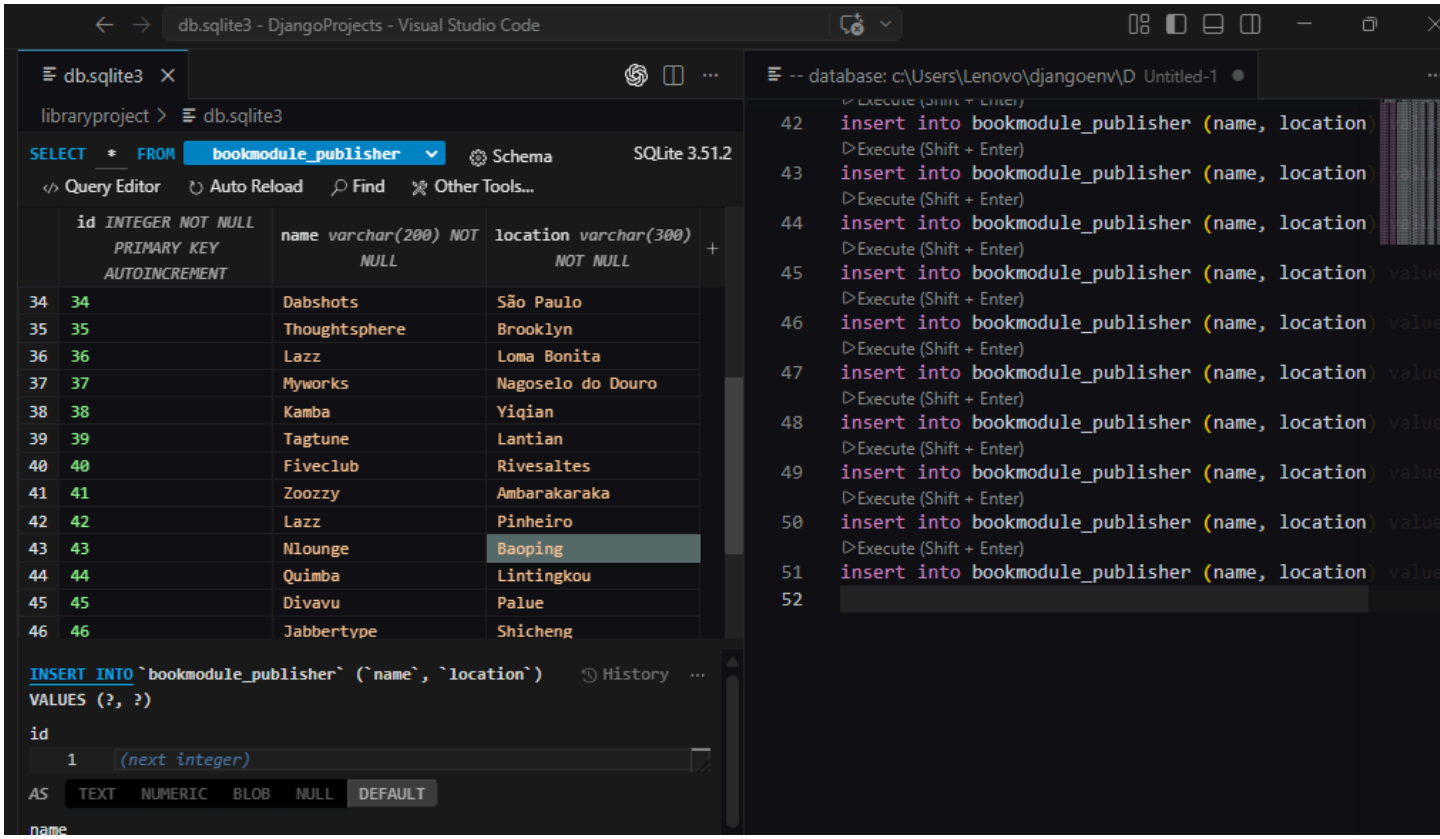
models.py

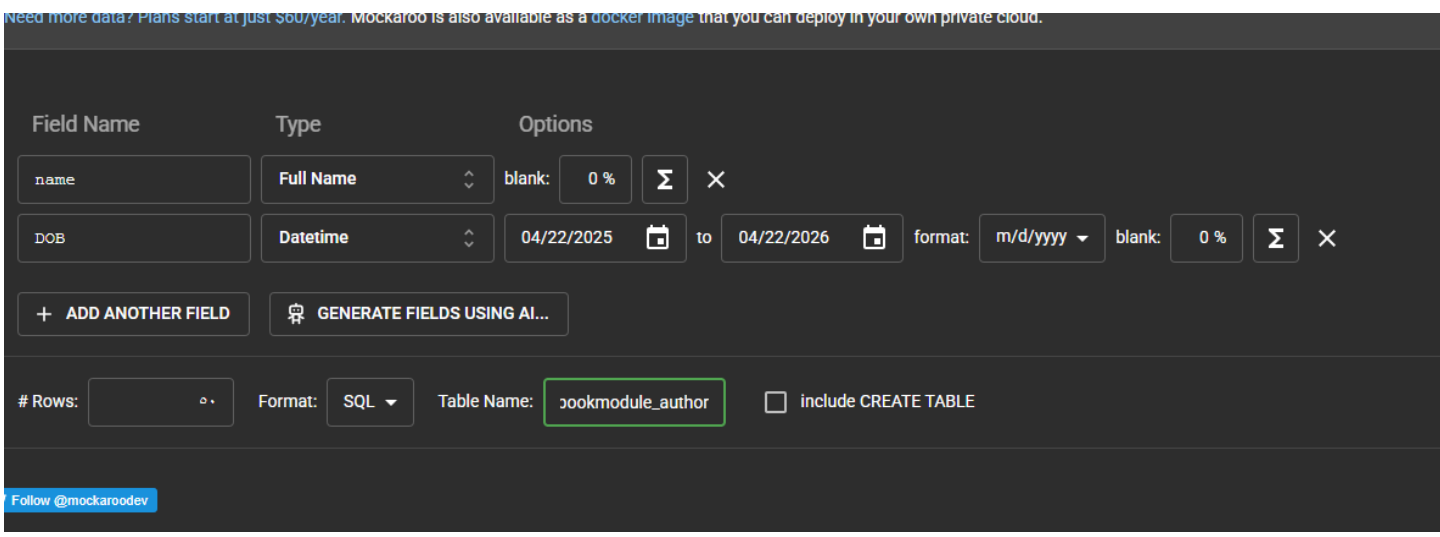
C: > Users > Lenovo > djangoenv > DjangoProjects > libraryproject > apps > bookmodule > models.py

```python

# Create your models here.

from django.db import models



from django.db import models

class Publisher(models.Model):
    name = models.CharField(max_length=200)
    location = models.CharField(max_length=300)

class Author(models.Model):
    name = models.CharField(max_length=200)
    DOB = models.DateField(null=True)

class Book(models.Model):
    title = models.CharField(max_length=100)
    price = models.FloatField(default=0.0)
    quantity = models.IntegerField(default=1)
    pubdate = models.DateTimeField()
    rating = models.SmallIntegerField(default=1)
    publisher = models.ForeignKey(Publisher, null=True, on_delete=models.SET_NULL)
    authors = models.ManyToManyField(Author)


class Address(models.Model):
    city = models.CharField(max_length=100)
```

2. Use fake data from mockaroo.com website or any other methods to fill tables above. Make sure that you handle the Book.publisher and Book.authors fields correctly.




db.sqlite3 - DjangoProjects - Visual Studio Code
db.sqlite3
libraryproject > db.sqlite3
SELECT * FROM bookmodule_publisher
Schema
SQLite 3.51.2
Query Editor
Auto Reload
Find
Other Tools...
id INTEGER NOT NULL PRIMARY KEY AUTOINCREMENT
name varchar(200) NOT NULL
location varchar(300) NOT NULL
34 34 Dabshots São Paulo
35 35 Thoughtsphere Brooklyn
36 36 Lazz Loma Bonita
37 37 Myworks Nagoselo do Douro
38 38 Kamba Yiqian
39 39 Tagtune Lantian
40 40 Fiveclub Rivesaltes
41 41 Zoozzy Ambarakaraka
42 42 Lazz Pinheiro
43 43 Nlounge Baoping
44 44 Quimba Lintingkou
45 45 Divavu Palue
46 46 Jabbertype Shicheng
INSERT INTO `bookmodule_publisher` (`name`, `location`)
VALUES (?, ?)
History
id
1 (next integer)
AS TEXT NUMERIC BLOB NULL DEFAULT
name
-- database: c:\Users\Lenovo\djangoenv\D Untitled-1
42 insert into bookmodule_publisher (name, location
43 insert into bookmodule_publisher (name, location
44 insert into bookmodule_publisher (name, location
45 insert into bookmodule_publisher (name, location
46 insert into bookmodule_publisher (name, location
47 insert into bookmodule_publisher (name, location
48 insert into bookmodule_publisher (name, location
49 insert into bookmodule_publisher (name, location
50 insert into bookmodule_publisher (name, location
51 insert into bookmodule_publisher (name, location
52
Execute (Shift + Enter)


Need more data? Plans start at just $60/year. Mockaroo is also available as a docker image that you can deploy in your own private cloud.
Field Name
Type
Options
name
Full Name
blank: 0 %
DOB
Datetime
04/22/2025
to
04/22/2026
format: m/d/yyyy
blank: 0 %
ADD ANOTHER FIELD
GENERATE FIELDS USING AI...
Rows:
Format: SQL
Table Name: bookmodule_author
include CREATE TABLE
Follow @mockaroodev

db.sqlite3

SELECT * FROM bookmodule_book | Schema | Query Editor | Auto Reload | Find | Other Tools...

	id INTEGER PRIMARY KEY	title TEXT	price REAL	quantity INTEGER	pubdate DATETIME
11	11	Programmer Analyst …	13.0	8	2025-12-08
12	12	Senior Developer	90.0	50	2026-02-12
13	13	Junior Executive	51.0	17	2026-02-22
14	14	Teacher	53.0	43	2025-07-15
15	15	Design Engineer	92.0	30	2025-06-26
16	16	Food Chemist	14.0	97	2025-07-20
17	17	Analyst Programmer	67.0	9	2025-05-30
18	18	Nurse	70.0	75	2026-03-23
19	19	Legal Assistant	54.0	16	2026-04-11
20	20	Editor	98.0	37	2025-07-13
21	21	Business Systems De…	74.0	39	2025-05-03
22	22	Food Chemist	13.0	30	2025-07-03
23	23	Office Assistant I	77.0	3	2025-07-09
24	24	Database Administra…	10.0	32	2025-11-07
25	25	Tax Accountant	90.0	7	2025-11-19
26	26	Environmental Tech	4.0	7	2025-09-15
27	27	Geological Engineer	40.0	73	2025-08-19
28	28	Help Desk Technician	99.0	7	2025-09-20
29	29	Health Coach I	63.0	17	2025-07-15
30	30	Software Engineer I…	15.0	28	2026-02-05

1 08-05-2025 AS TEXT NUMERIC BLOB

Field Name	Type	Options
title	Job Title	blank: 0 %
price	Number	min: ١ max: ١٠٠ decimals: ٠ blank: 0 %
quantity	Number	min: ١ max: ١٠٠ decimals: ٠ blank: 0 %
pubdate	Datetime	04/22/2025 to 04/22/2026 format: m/d/yyyy blank: 0 %
rating	Row Number	blank: 0 %

+ ADD ANOTHER FIELD | GENERATE FIELDS USING AI...

Rows: ٥٠ Format: SQL Table Name: bookmodule_book ☐ include CREATE TABLE

Lab Activities:

Task 1: Create a URL '/books/lab9/task1' with any necessary HTML file and view function to list books along with percentage availability. For example, if the number of books in 350 and the quantity of a book is 7, then the percentage of availability is 2%. Use transient field for the percentage.

```
def lab9_task1(request):
    total_stock_data = Book.objects.aggregate(total=Sum('quantity'))
    total_stock = total_stock_data['total'] or 1

    books = Book.objects.filter(price__gt=50).annotate(
        availability_priority=ExpressionWrapper(
            (F('quantity') * 100.0) / Value(total_stock),
            output_field=FloatField()
        )
    )
    return render(request, 'bookmodule/lab9_task1.html', {'books': books})
```

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Books with Price > 50 and Availability Priority

Title	Price	Quantity	Availability Priority
Senior Editor	99.0	12	0.59%
Office Assistant IV	97.0	63	3.11%
Research Assistant IV	77.0	13	0.64%
Payment Adjustment Coordinator	51.0	2	0.10%
Analyst Programmer	53.0	84	4.15%
Help Desk Technician	96.0	70	3.46%
Graphic Designer	95.0	85	4.20%
Senior Developer	90.0	50	2.47%
Junior Executive	51.0	17	0.84%
Teacher	53.0	43	2.12%
Design Engineer	92.0	30	1.48%
Analyst Programmer	67.0	9	0.44%
Nurse	70.0	75	3.70%
Legal Assistant	54.0	16	0.79%
Editor	98.0	37	1.83%
Business Systems Development Analyst	74.0	39	1.93%
Office Assistant I	77.0	3	0.15%
Tax Accountant	90.0	7	0.35%
Help Desk Technician	99.0	7	0.35%

Task 2: Create a URL '/books/lab9/task2', HTML file, and query list books that all publishers, annotated with their respective total book stock.

```python
def lab9_task2(request):
    publishers = Publisher.objects.annotate(
        total_stock=Sum('book__quantity'),
        num_books=Count('book')
    )
    return render(request, 'bookmodule/lab9_task2.html', {'publishers': publishers})
```

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Publisher Insights (Total Stock)

List of publishers with their respective total book stock.

Publisher Name	Total Book Stock
Bubbletube	1153
Yakitri	872
Quinu	0
Einti	0

Task 3: (opposite to Task 2) Create a URL '/books/lab9/task3', HTML file, and a query to get oldest book of any of those managed by every publisher.

127.0.0.1:8000/books/lab9/task3

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Publisher Insights (Oldest Books)

- **Publisher:** Bubbletube
  Total Books: 30
  Oldest Book Date: 2025-04-23
- **Publisher:** Yakitri
  Total Books: 20
  Oldest Book Date: 2025-04-22

Task 4: Create a URL '/books/lab9/task4', HTML file, and query to calculate the average, min, and max price of books for each publisher.

```python
def lab9_task4(request):
    publishers = Publisher.objects.annotate(
        avg_price=Avg('book__price'),
        min_price=Min('book__price'),
        max_price=Max('book__price')
    )
    return render(request, 'bookmodule/lab9_task4.html', {'publishers': publishers})
```

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Publisher Price Statistics

Publisher Name	Average Price	Minimum Price	Maximum Price
Bubbletube	76.87	51.0	99.0
Yakitri	19.70	1.0	50.0

Task 5: Create a URL '/books/lab9/task5', HTML file, and a query to list publishers with a count of highly rated books, with their names and quantity.

```
from django.db.models import Count, Q
from .models import Publisher
def lab9_task5(request):
    publishers = Publisher.objects.annotate(
        highly_rated_count=Count(
            'book',
            filter=Q(book__rating__gte=4)
        )
    )
    return render(request, 'bookmodule/lab9_task5.html', {'publishers': publishers})
```

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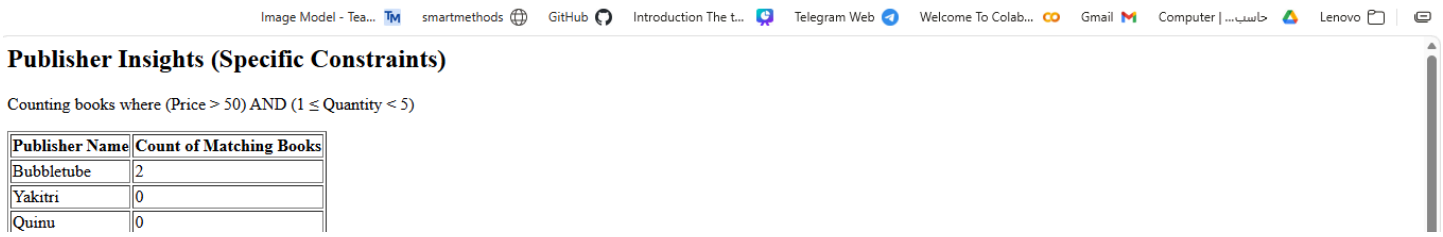
Publishers & Highly Rated Books

List of publishers with the count of books rated 4 or higher.

Publisher Name	Quantity of Highly Rated Books
Bubbletube	29
Yakitri	18
Quinu	0
Einti	0
Meeveo	0
Photobean	0
Jaloo	0
Skinte	0
Tagcat	0
Zoombeat	0

Task 6: Create a URL '/books/lab9/task6', HTML file, and a query to count of books for each publisher, filter only books with price greater than 50 and quantity less than 5 and greater than or equal to 1.

```python
def lab9_task6(request):
    publishers = Publisher.objects.annotate(
        filtered_books_count=Count(
            'book',
            filter=Q(book__price__gt=50) & Q(book__quantity__lt=5) & Q(book__quantity__gte=1)
        )
    )
    return render(request, 'bookmodule/lab9_task6.html', {'publishers': publishers})
```